

## Biology — Exam 02 — Chapter 7: Metabolism

Seemab | Annual Examination 2026 | Federal Board (FBISE) SSC-I | Attempt date: 30 April 2026 | Marked: 30 April 2026

TOTAL SCORE

60.5 / 65

PERCENTAGE

93.1%

GRADE

A

MCQ SCORE

12 / 12

### SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks) Score: 12/12

| #  | Question (short)                                                | Seemab                         | Key                            | Result |
|----|-----------------------------------------------------------------|--------------------------------|--------------------------------|--------|
| 1  | Sum of all chemical reactions in a cell                         | C — Metabolism                 | C — Metabolism                 | 1/1    |
| 2  | Enzymes are produced by                                         | B — Living cells               | B — Living cells               | 1/1    |
| 3  | Part of enzyme where substrate binds                            | C — Active site                | C — Active site                | 1/1    |
| 4  | Enzyme loses function at high temperature because it is         | B — Denatured                  | B — Denatured                  | 1/1    |
| 5  | Competitive inhibitor blocks enzyme by binding to               | C — Active site                | C — Active site                | 1/1    |
| 6  | ATP is energy currency because it                               | B — Stores and releases energy | B — Stores and releases energy | 1/1    |
| 7  | Third component of ATP alongside adenine and ribose             | B — Three phosphate groups     | B — Three phosphate groups     | 1/1    |
| 8  | O <sub>2</sub> released as by-product of photosynthesis through | B — Photolysis of water        | B — Photolysis of water        | 1/1    |
| 9  | Calvin cycle occurs in the                                      | C — Stroma of chloroplast      | C — Stroma of chloroplast      | 1/1    |
| 10 | Net ATP gain from glycolysis                                    | C — 2 ATP                      | C — 2 ATP                      | 1/1    |
| 11 | Alcoholic fermentation in yeast produces ethyl alcohol and      | C — Carbon dioxide             | C — Carbon dioxide             | 1/1    |
| 12 | Final electron acceptor in ETC                                  | C — Oxygen                     | C — Oxygen                     | 1/1    |

Seven consecutive perfect 12/12 MCQ scores — record-extending streak continues.

### SECTION B — Short Questions (11 × 3 = 33 Marks) Score: 30.5/33

#### Q.2(i) MAIN — Define metabolism; differentiate catabolism and anabolism with one example each

3/3

##### Main question

Metabolism defined correctly ("sum of all chemical reactions going on within each cell ... for the maintenance and sustainability of life"). Catabolism defined with digestion example. Anabolism defined with glycogen synthesis example. All three mark-points hit cleanly.

Metabolism def: 1/1 Catabolism + example: 1/1 Anabolism + example: 1/1

#### Q.2(ii) OR — Explain "enzymes are specific"; give one example

2/3

##### OR variant chosen

Specificity concept stated well: "enzymes are usually very specific so catalyse only one type of reaction." Amylase example given (acts only on starch, not proteins or fats).

Missing the second mark-point: the answer key requires an explicit statement that "the active site has a unique shape fitting only one specific substrate — like a lock fits only one key." The mechanism behind specificity (active-site shape) was not mentioned. Lost 1 mark.

Specificity concept: 1/1 Active site mechanism: 0/1 Example (amylase): 1/1

#### Q.2(iii) OR — What is energy of activation? How do enzymes lower it?

3/3

##### OR variant chosen

Energy of activation defined precisely ("minimum amount of extra energy required to activate the molecule to react or start the reaction"). Both mechanisms stated: (1) enzymes lower it by changing the shape and charge of the substrate; (2) enzymes may provide an entirely different mechanism of reaction. Body-temperature significance included.

Definition: 1/1 How enzymes lower it (2 mechanisms): 1/1 Effect / body temp: 1/1

#### Q.2(iv) MAIN — How does temperature affect enzyme activity? Very high and very low?

3/3

##### Main question

High temperature: atoms vibrate vigorously → globular structure and active site damaged → denatured. Low temperature: enzyme becomes inactive due to fall in available activation energy. Both effects clearly stated with correct terminology.

High temp / denaturation: 1/1 Low temp / inactivity: 1/1 Optimum context implicit: 1/1

#### Q.2(v) MAIN — Distinguish competitive and non-competitive inhibition

2/3

##### Main question

Competitive inhibition: binding site (active site) correctly identified; shape similarity to substrate stated ("molecules which occupy the active site due to their similarity in shape or structure to substrate"). Reversibility contrast present (can be reversed / cannot be reversed).

Allosteric site not named — for non-competitive inhibition, the answer key requires explicitly naming the allosteric site as the binding location, not just stating that it doesn't bind the active site. Seemab wrote "do not occupy the active site" but did not name the allosteric site. Lost 1 mark.

Binding site contrast (active vs allosteric): 1/1 Shape similarity to substrate (stated): 1/1 Allosteric site named: 0/1

**Q.2(vi) MAIN — Describe structure of ATP molecule; name its three components****3/3****Main question**

All three components named correctly: Adenine (double-ringed nitrogen base), Ribose (five-carbon sugar), Three phosphate groups (PO<sub>4</sub>) in triphosphate chain. Adenosine formation explained (Adenine + 5C sugar → Adenosine). AMP → ADP → ATP chain described with phosphate additions. Excellent structural precision.

Adenine: 1/1   Ribose: 1/1   Three phosphate groups: 1/1

**Q.2(vii) MAIN — Overall equation for photosynthesis; raw materials and products****3/3****Main question**

Equation written correctly:  $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow (\text{light/chlorophyll}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$ . Raw materials (CO<sub>2</sub> from stomata, H<sub>2</sub>O absorbed by roots, chlorophyll captures light), products (glucose + oxygen), and conditions (light energy + chlorophyll, plants/algae/cyanobacteria) all stated.

Equation: 1/1   Raw materials: 1/1   Products + conditions: 1/1

**Q.2(viii) OR — Describe photolysis of water; its role in light-dependent reactions****3/3****OR variant chosen**

Definition: light-dependent reactions depend on Photosystems I and II to absorb light. Role I: electrons released by photolysis replace electrons lost from excited chlorophyll of PSII — maintaining the reaction. Role II: O<sub>2</sub> released through stomata, H<sup>+</sup> ions reduce NADP<sup>+</sup> to NADPH. All three mark-points present.

Photolysis definition: 1/1   Role I (electron replacement): 1/1   Role II (O<sub>2</sub> + NADPH production): 1/1

**Q.2(ix) MAIN — Four stages of aerobic respiration in correct order****2.5/3****Main question**

All four stages named in correct order: Glycolysis (cytoplasm), Formation of Acetyl CoA (mitochondria), Krebs Cycle, Electron Transport Chain. Equations for each stage written out correctly with products.

ETC location not stated — answer key requires "inner mitochondrial membrane." The answer says only " $\text{NADH} + \text{FADH}_2 \rightarrow 32 \text{ATP} + \text{H}_2\text{O}$ " without specifying the location. Partial mark for ETC (0.5 mark deducted; location mark not fully awarded).

4 stages named in order: 1/1   Locations (glycolysis/mitochondria): 1/1 — but ETC location missing   Products per stage: 0.5/1

**Q.2(x) MAIN — What is glycolysis? Where does it occur? Products?****3/3****Main question**

Definition precise: "first stage of cellular respiration, one glucose (6C) broken down into two molecules of pyruvic acid (3C each)." Location: cytoplasm, outside mitochondria. Products: 2 pyruvic acid + 2 ATP (net) + 2 NADH — "(net)" specifically included, showing understanding of gross vs net yield.

Definition: 1/1   Location: 1/1   Products (with net qualifier): 1/1

**Q.2(xi) OR — Four reasons why anaerobic respiration is important****3/3****OR variant chosen**

All four reasons given: (1) Early Earth / earliest organisms — anaerobic respiration was evolutionary necessity. (2) Bacteria use it for cheese and yogurt making. (3) Yeast fermentation for wine making and baking industry. (4) Lactic acid fermentation provides emergency energy to muscle cells during vigorous exercise when O<sub>2</sub> is insufficient. Matches answer key exactly.

Reason 1 (evolutionary): 1 mark   Reason 2 (food industry): 1 mark   Reasons 3+4: 1 mark

**SECTION C — Long Questions (4 × 5 = 20 Marks)   Score: 18/20****Q.3(i) MAIN — Define enzyme; describe any five characteristics with examples****4/5****Main question**

Enzyme defined excellently: "biological catalysts which are biologically active globular proteins made by living cells, which speed up extremely biochemical reaction (without being used)." Five characteristics written: (1) Globular proteins — amino acid chains folded into 3D globular proteins, ribozymes mentioned. (2) Specific — with amylase example. (3) Required in small quantity — reuse mechanism explained. (4) Regulation of enzyme production and activity — synthesis can increase/decrease by cell; inhibitors and activators. (5) Enzymes make metabolic pathways — series of connected chemical reactions described.

Characteristic 2 (Increase rate of reaction) — the key expects "millions of times faster than non-enzymatic reactions; only speed up, do not start reactions." Seemab's answer does not include "increase rate of reaction" as a named characteristic with that specific quantitative claim. The specificity and metabolic pathway characteristics are valid but the "rate of reaction" characteristic earns its own mark. Lost 1 mark for omitting the speed/rate-of-reaction characteristic as a distinct point.

Definition: 1/1   Globular proteins: 1/1   Rate of reaction (missing explicitly): 0/1   Small quantity + reuse: 1/1   Specific + example: 1/1   Regulation/metabolic pathway: 1/1 (takes "also valid" credit)

**Q.3(ii) OR — Competitive and non-competitive inhibition; allosteric site; daily-life examples****5/5****OR variant chosen**

Enzyme inhibition intro: definition, inhibitor binds in place of substrate, does not convert to products, external factors listed (poisons, cyanide, antibiotics, drugs). Competitive inhibition: inhibitor resembles substrate → occupies active site → substrate blocked → no product → reversible → sulfonamide drug example with antibiotic action. Non-competitive inhibition: does not resemble substrate → binds allosteric site (enzyme surface, not active site) → alters globular shape → active site damaged → substrate cannot bind → irreversible → cyanide (rat poison) inhibiting cytochrome c oxidase in ETC → heavy metals (Pb, Hg) also mentioned. All five mark-points hit.

Inhibition intro: 1/1   Competitive (mechanism + reversible + example): 2/2   Non-competitive (allosteric + irreversible + example): 2/2

OR variant chosen

Overall equation written. Light-dependent reactions (Phase 1) described in sequence with location (thylakoid membranes, grana), starting at PS II, chlorophyll absorbs light, photolysis of water ( $\text{H}_2\text{O} \rightarrow \text{O}_2 + 2\text{H}^+ + \text{electrons}$ ), electrons pass through ETC for ATP synthesis, PS I reduces  $\text{NADP}^+$  to NADPH. Products: ATP, NADPH. Calvin cycle (Phase 2) in stroma of chloroplast:  $\text{CO}_2 + \text{RuBP (5C)} \rightarrow \text{6C intermediate} \rightarrow 2 \times 3\text{-PGA} \rightarrow \text{reduced by NADPH} + \text{ATP} \rightarrow \text{G3P (3C sugar)} \rightarrow \text{glucose synthesis} + \text{RuBP regeneration}$ . Melvin Calvin credited.

Overall balanced chemical equation  $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$  not written explicitly in Q3(iii). Raw materials and products were stated in prose but the balanced equation itself was missing. (Note: Seemab DID include "O<sub>2</sub> released through stomata" in the Phase 1 photolysis step — that sub-point is present and credited.) Lost 1 mark.

Overall balanced equation (not written): 0/1   Phase 1 — light reactions (complete): 2/2   Phase 2 — Calvin cycle: 2/2

Q.3(iv) MAIN — Compare aerobic and anaerobic respiration; glycolysis as common stage; fate of pyruvic acid

Main question

Comparison table: oxygen needed (yes/no), glucose oxidation (complete  $\rightarrow \text{CO}_2 + \text{H}_2\text{O}$  vs incomplete  $\rightarrow \text{alcohol/lactic acid}$ ), ATP yield (36 vs 2 ATP), site (cytoplasm + mitochondria vs cytoplasm only) — all four features correct. Glycolysis as common first stage: both pathways start with glycolysis in cytoplasm, Glucose (6C)  $\rightarrow$  2 pyruvic acid (3C) + 2 ATP + 2 NADH. Fate in anaerobic: pyruvic acid not fully oxidised — yeast  $\rightarrow$  ethyl alcohol +  $\text{CO}_2$  (wine/bread), muscles  $\rightarrow$  lactic acid (yogurt, exercise fatigue). Fate in aerobic: pyruvic acid enters mitochondria  $\rightarrow$  Step 2 Acetyl CoA  $\rightarrow$  Step 3 Krebs  $\rightarrow$  Step 4 ETC, total 36 ATP with  $\text{O}_2$  as final electron acceptor. All five mark-points present — full marks awarded. (The overall aerobic equation  $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$  is only required in the OR variant, not the main question.)

Comparison table (4 features): 1/1   Glycolysis as common stage: 1/1   Fate in anaerobic (yeast + muscle examples): 1/1   Fate in aerobic (all 3 stages + 36 ATP): 1/1   All required elements present: 1/1

**Score reconciliation:** Section A: 12/12 | Section B: Q2(i) 3 + Q2(ii)or 2 + Q2(iii)or 3 + Q2(iv) 3 + Q2(v) 2 + Q2(vi) 3 + Q2(vii) 3 + Q2(viii)or 3 + Q2(ix) 2.5 + Q2(x) 3 + Q2(xi)or 3 = **30.5/33** | Section C: Q3(i) 4 + Q3(ii)or 5 + Q3(iii)or 4 + Q3(iv) 5 = **18/20** | **Total: 60.5/65 (93.1%)**

Summary — Strengths and Areas for Improvement

Strengths

- Seventh consecutive perfect 12/12 MCQ score — streak extended to 7 in a row
- ATP structure described with exceptional precision (adenosine formation,  $\text{AMP} \rightarrow \text{ADP} \rightarrow \text{ATP}$  chain)
- Anaerobic respiration importance — all 4 reasons correctly stated from memory
- Glycolysis answer: "(net)" qualifier spontaneously used — shows understanding of gross vs net ATP
- Section C Q3(ii) inhibition answer: both competitive and non-competitive with examples from daily life (sulfonamide, cyanide, heavy metals) — textbook-level completeness
- Four-stage aerobic respiration: all equations written correctly with correct products at each stage
- Calvin cycle steps written correctly in sequence with names (RuBP, 3-PGA, G3P, Melvin Calvin)
- Q3(iv) — full 5/5 on aerobic vs anaerobic comparison: all 4 table features, glycolysis as common stage, both anaerobic fates (yeast + muscle), all 3 aerobic stages with 36 ATP total
- Massive improvement vs Biology exam-01 (52/65, 80%) — jumped 13.1 percentage points to 93.1%

Areas for Improvement

- Q2(ii) OR — Enzyme specificity mechanism missing:** Stated that enzymes are specific and gave amylase example, but did not explain WHY (active site has unique shape — lock-and-key). Always link specificity to active-site geometry.
- Q2(v) — Allosteric site not named:** For non-competitive inhibition, the answer key requires explicitly naming the allosteric site as the binding location. Writing "do not occupy the active site" is necessary but not sufficient — the allosteric site must be named directly.
- Q2(ix) — ETC location not stated:** "Inner mitochondrial membrane" is a required mark-point for locating ETC. Named all 4 stages but forgot to place ETC on the inner membrane.
- Q3(i) — "Increase rate of reaction" not named explicitly:** Listed 5 valid characteristics but omitted the speed/rate characteristic as a named, distinct point. The quantitative claim ("millions of times faster") earns its own mark.
- Q3(iii) — Overall balanced equation not written:**  $6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{H}_2\text{O} + 6\text{O}_2$  was not written explicitly in the long answer. Raw materials and products were stated in prose — the balanced equation itself must appear. (Note: Q2(vii) MAIN had the equation correct; Q3(iii) OR did not include it.)

Comparison — Biology Exam Performance

| Exam          | Date        | Score   | %     | MCQ   | Sec B   | Sec C  | Notes                                              |
|---------------|-------------|---------|-------|-------|---------|--------|----------------------------------------------------|
| exam-01-ch1-6 | 3 Apr 2026  | 52/65   | 80.0% | 11/12 | ~26/33  | ~15/20 | Q.5 Section C not attempted (-5); time issues      |
| exam-02-ch7   | 30 Apr 2026 | 60.5/65 | 93.1% | 12/12 | 30.5/33 | 18/20  | All sections complete; 7th consecutive perfect MCQ |

**+13.1 percentage points improvement vs Biology exam-01.** The time-management issue (unattempted Q.5) from exam-01 is fully resolved — all 4 Section C questions answered. MCQ error from exam-01 (missed Q5) not repeated.